

Hashbrown Password Vault

A local, encrypted password manager with a graphical interface. Each user's credentials are stored in their own encrypted vault file, protected by their master password.

Tools & Technologies

- **Python 3.11+**
- **customtkinter** — GUI framework
- **SQLite** — Unencrypted user account database (users.db)
- **SQLCipher** — Encrypted per-user vault databases
- **Argon2** — Master password hashing
- **AES-256-GCM** — Individual credential encryption

Requirements

Install all dependencies with:

```
pip install -r requirements.txt
```

requirements.txt

```
argon2-cffi==25.1.0
argon2-cffi-bindings==25.1.0
cffi==2.0.0
cryptography==46.0.7
customtkinter==5.2.2
darkdetect==0.8.0
packaging==26.0
pillow==12.2.0
pyparser==3.0
sqlcipher3==0.6.2
```

Note: Developed and tested on Linux Mint. If running on Windows, WSL is recommended due to native C library dependencies in sqlcipher3 and argon2-cffi.

Running the App

```
python3 main.py
```

The application will automatically create users.db and the vaults/ directory on first launch. No manual database setup is required.

Project Structure

```
Hashbrown Password Vault/
├── main.py
├── requirements.txt
├── utils.py
├── database/
│   ├── db.py
│   ├── crypto.py
│   ├── test_crypto.py
│   ├── test_db.py
│   └── users.db          (auto-generated)
├── screens/
│   ├── login_screen.py
│   ├── create_account_screen.py
│   ├── main_menu.py
│   ├── vault_screen.py
│   ├── pw_generator.py
│   └── settings_screen.py
├── vaults/
│   └── <username>.db    (auto-generated per user)
```